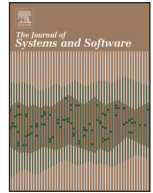




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Editorial

Introduction to the special issue on “New frontiers in software architecture”

Introduction

Since the early 1990s, software architecture has been subject of increasing interest in software engineering research and practice. Three classic reasons why architecture is important to software systems are: (1) architecture provides a basis for creating mutual understanding about the software system; (2) architecture design decisions have the most significant influence on system qualities; and (3) architecture provides a reusable abstraction of a system that is transferable to other systems with similar requirements promoting large-scale reuse. In the last decade, the awareness has grown that the decision making process is central to architectural design, as manifested in the notions of rational and architecture knowledge.

Despite the maturity of the field, software architecture does not escape from the impact of the ever-changing world in which we live. The need for on the fly re-configurability and self-adaptation to deal with dynamic operating conditions and new emerging user requirements poses severe challenges to software architects. New developments such as the cloud, virtualization, big data, ultra-large scale systems, long-lived systems and systems-of-systems require new architecture solutions. The aim of this special issue is to document new frontiers in software architecture and outline the challenges for the coming years.

This issue

The idea for this special issue arose from conversations with the organizing committee of the Ninth European Conference on Software Architecture (ECSA), which was held in Cavtat/Dubrovnik, Croatia, September 2015. The response to the call for papers for this special issue drew heavily from the ECSA community, but also attracted submissions more broadly. We received seventeen submissions for this special issue. These papers were rigorously reviewed by an international group of experts in the field of software architectures. Based on these reviews, we were able to recommend the following six papers for publication.

- “A Hybrid and Learning Agent Architecture for Network Intrusion Detection,” by Adriana Leite and Rosario Girardi, proposes a software agent architecture that combines case-based reasoning, reactive behavior, and learning. The effectiveness and performance of the agent architecture is evaluated in four experi-

ments to detect intrusions in dynamic environments using different combinations of behavior with and without learning.

- “Designing and Applying an Approach to Software Architecting in Agile Projects in Education”, by Samuil Angelov and Patrick de Beer, presents an approach to the introduction of software architecting activities in agile projects in education. In particular, the article illustrates the experiences and lessons learned from two applications of the approach and offers research directions for improving education on architecting in agile projects.
- “Exploiting Traceability Uncertainty between Software Architectural Models and Extra-Functional Results” by Catia Trubiani, Achraf Ghabi, and Alexander Egyed, introduces an approach to automatically generate traceability links between architectural model elements and extra-functional results. Traceability uncertainty is mitigated by using extra-functional property patterns and anti-patterns. The approach is validated in two academic projects and an industrial system.
- “A Search Engine for Finding and Reusing Architecturally Significant Code” by Ibrahim Jameel Mujhid, Joanna C. Santos, Raghuram Gopalakrishnan, and Mehdi Mirakhorli presents a code search engine to automate the discovery, extraction, and indexing of tactics. In particular, novel techniques to detect tactics and technical context in which they are used are introduced, as well as a novel ranking algorithm to order the retrieved tactical files.
- “A Domain-specific Language for The Control of Self-adaptive Component-based Architecture” by Frederico Alvares, Eric Rutten, and Lionel Seinturier, proposes a high-level language - Ctrl-F - for the design of self-adaptive software components based on logical discrete control approaches, in which the self-adaptive behavioral models enrich component controllers with a knowledge of events, configurations and past history, and possible future configurations. Ctrl-F has been integrated with FraSCAti, a Service Component Architecture middleware platform and its usefulness has been illustrated using two case studies.
- “Generating Reusable, Searchable and Executable Architecture Constraints as Services” by Sahar Kallel, Bastien Tramoni, Chouki Tibermacine, Christophe Dony, and Ahmed Hadj Kacem, introduces a process for generating reusable and executable architecture constraints by automatically creating components that embed architecture constraints and deploying the constraints as services published in a registry. The effectiveness of

the proposed process has been experimented on a set of 15 architecture constraints and in a real-world system.

We hope that these papers will help provide the community with an understanding of new frontiers in software architecture and challenges for the coming years.

Acknowledgments

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